Working Group Rusestky: Updates

Ajay S. Sakthivasan[1]

Akaki Rusestky[1], Rishabh Bubna[1],
Ming-Yang Duan[2], Kakha Shamanauri[1] et al.

[1]HISKP & BCTP, University of Bonn
[2]UCAS, Beijing

September 11, 2025

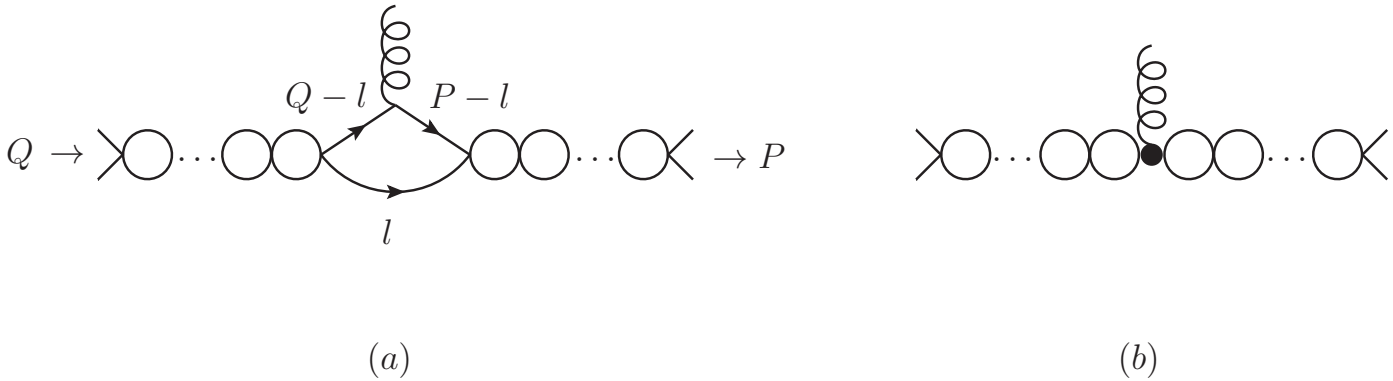
The Electromagnetic Form Factor of the ρ-meson on the Lattice

Ajay Sakthivasan et al.

Form Factor Fundamentals

- The calculation of the form factors of unstable states on the lattice pose significant challenges.
- In this work, we discuss the conceptual challenges associated with the calculation of the ρ-meson electromagnetic form factor.
- Given a quantum field with the quantum numbers of the ρ-meson obeying relevant constraints:
  - $\rightsquigarrow$ The two-point function develops a simple pole on the second Riemann sheet at $s = s_R$.
  - $\rightsquigarrow$ The three-point function involving the electromagnetic current develops a double pole at $s = s' = s_R$.
  - $\rightsquigarrow$ The extraction of the EM form factor corresponds to calculating the residue of the three-point function.

- The contributions can be split into a triangle diagram and the contact term:



- The triangle diagram does not exhibit a regular dependence on the box size L. (*Bernard et al., 2012*)
- To overcome this, we make use of the external field method. (*Lozano et al., 2022*).
  - ⇝ The system is placed in a periodic external field, which is coupled to a given current.
  - ⇝ The modified Lüscher approach lets one extract relevant physical results.

- The objective is to relate the quantities extracted from the lattice to their infinite volume counterparts in a consistent way.
- Provide numerical estimates for the quantities that are to be extracted from the lattice.

The form factor in the NREFT framework

- We work with an NREFT Lagrangian that describes the elastic $\pi^+\pi^0$ scattering with a dipion field Ψ that models the ρ-meson:

$$\mathcal{L} = \overbrace{\sum_{a=+,0} \phi_a^\dagger 2W(i(v\cdot\partial) - W)\phi_a}^{\text{Pion Lagrangian}} - \overbrace{\Psi_\mu^\dagger \Psi^\mu}^{\text{Rho Lagrangian}} .$$

$$W = \sqrt{M^2 + \partial^2 - (v\cdot\partial)^2} \quad \text{quantisation axis}$$

$$\Psi^\mu = \phi_+ \overset{\leftrightarrow}{W}{}^\mu f(\overset{\leftrightarrow}{W}{}^2)\phi_0,$$
$$W^\mu = v^\mu W + i(\partial^\mu - v^\mu (v\cdot\partial)),$$
$$f(\overset{\leftrightarrow}{W}{}^2) = f_0 + f_1 \overset{\leftrightarrow}{W}{}^2 + \ldots.$$

- f is related to the $\pi\pi$ scattering phase shift in the p-wave:

$$(q_0 \cot\delta_1(q_0))^{-1} = -\frac{q_0^2 f^2(q_0^2)}{24\pi\sqrt{P^2}}$$

Inclusion of Electromagnetic Interaction

- To study the electromagnetic form factor, we need to gauge the NREFT Lagrangian.
- This is not straightforward due to the presence of non-local operators.
- Under minimal coupling, we have

$$W^2 = M^2 + \partial^2 - (v \cdot \partial)^2 \to W^2 + B + \mathcal{O}(e^2).$$

Then it can be shown that

$$W \to W + D, \quad D = \int_0^\infty dt \; e^{-tW} B e^{-tW},$$

which can be evaluated in the momentum space.

- Gauging the four-pion interaction Lagrangian yields local terms, which we manually include:

$$\mathcal{L}_{\text{c.t.}} = \sum_{i=1}^{3} g_i e \mathscr{O}_i,$$

where

$$\mathscr{O}_1 = \Psi_\mu^\dagger \{\partial^\beta F_{\alpha\beta}, W_\alpha\} g^{\mu\nu} \Psi_\nu,$$
$$\mathscr{O}_2 = \Psi_\mu^\dagger i F^{\mu\nu} \Psi_\nu,$$
$$\mathscr{O}_3 = \Psi_\mu^\dagger [W^\mu, [W^\nu, \{\partial^\beta F_{\alpha\beta}, W_\alpha\}]] \Psi_\nu.$$

- The matrix element of the electromagnetic current between the two-pion states can be evaluated, and it is parametrised in terms of the couplings g_1, g_2 and g_3.

Matching of the Couplings to ChPT Amplitude

- The couplings g_1, g_2 and g_3 can be matched to $\mathcal{O}(p^4)$ ChPT LEC's, which yields:

$$eg_2 f^2(0) = \frac{2e(2l_1^r - l_2^r)}{F_\pi^4} + \text{loops},$$

$$eg_1 f^2(0) = eg_3 f^2(0) = 0 + \text{loops}.$$

- We make use of Schenk parametrisation for the phase shift, which yields

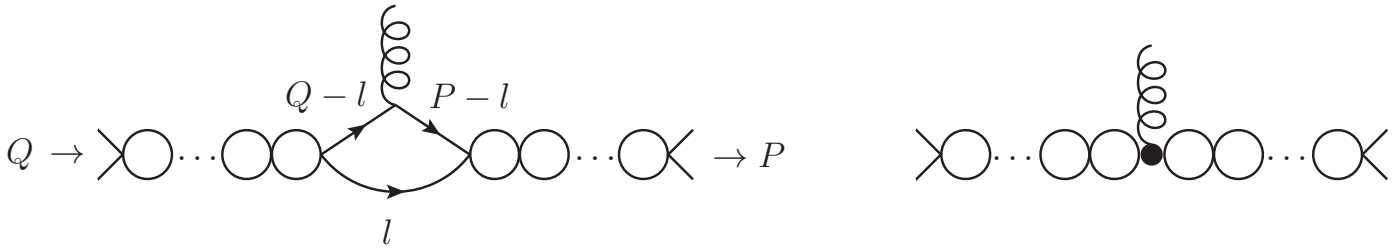
$$g_2 = -\frac{M_\pi^2(2l_1^r - l_2^r)}{24\pi F_\pi^4 a_1^1} + \text{loops} = 2.8 \cdot 10^{-2} M_\pi^{-2},$$

$$g_1 = g_3 = 0 + \text{loops}.$$

- We assume that the loop contributions are of the similar order.
- This matching gives a numerical estimate of the couplings—which are regular and can be extracted from lattice.
- A detailed numerical analysis of the matching process and an alternate estimate of the couplings g_1, g_2 and g_3 based on the explicit inclusion of a ρ-meson field in ChPT is also in progress. (*Shamanauri et al.*)

The Form Factor in the Infinite Volume

- As mentioned before, the contributions to the form factor are from a triangle diagram and the contact term.



(a) (b)

- The contributions can now be calculated in the NREFT and related to the invariant form factor parameters.

Lüscher Equation in the External Field

- Placing the system in a periodic electromagnetic field on a cubic lattice allows one to extract the invariant form factors.
- For an unstable particle, the energy levels are determined by the modified Lüscher approach with the quantisation condition:

$$\det(\Pi^{-1}_{\mu\nu} - D^{0}_{\mu\nu}) = 0$$

"self-energy" the two-point function of the interpolating field

- Note that the interpolating field is related to the dipion field Ψ.

- The couplings g_1, g_2 and g_3 can be evaluated by projecting on to the relevant irreps.
- Use g_1, g_2 and g_3 to evaluate the infinite volume invariant form factors (numerical analysis in progress).
- Also, obtain numerical estimates of g_1, g_2 and g_3 starting from infinite volume form factors.

Modified Lüscher Zeta-function and the Modified Effective Range Expansion in the Presence of a Long-range Force

Rishabh Bubna et al.

Long-range forces on the lattice (NN scattering)

Exchange of a light particle: pion

- Partial-wave mixing is sizable in the quantisation condition.
  $\rightarrow$ *Slowly converging partial-wave expansion* (Meng & Epelbaum, 2021).

- Some energy levels are located on the t-channel cut ($-\infty < s \leq (2m_N)^2 - M_\pi^2$), close to threshold.
  $\rightarrow$ *Standard Lüscher approach not applicable*.

- Exponentially suppressed correction are sizable.
  $\rightarrow$ $e^{-M_\pi L}$ *not suppressed*.

Modified Lüscher equation

- Define modified effective-range function (van Haeringen & Kok, 1982):

$$K_l^M(q^2) = M_l(q) + \frac{q^{2l+1}}{|f_l(q)|^2}(\cot(\delta_l(q) - \sigma_l(q)) - i)$$

- The radius of convergence of $K_l^M(q^2)$ is determined by the range of the short-range potential.
- Quantization condition:

$$\begin{aligned} \det \mathscr{A}_{lm,l'm'} &= 0, \\ \mathscr{A}_{lm,l'm'} &= \delta_{ll'}\delta_{mm'} K_l^M(q_0^2) - H_{lm,l'm'}(q_0). \end{aligned}$$

- Modified Lüscher zeta-function:

$$H_{lm,l'm'}(q_0) = \frac{4\pi}{L^6} \sum_{\boldsymbol{p},\boldsymbol{q}} \mathscr{Y}^*_{lm}(\boldsymbol{p}) \langle \boldsymbol{p}| G_L(q_0^2) |\boldsymbol{q}\rangle \mathscr{Y}_{l'm'}(\boldsymbol{q})$$

Set-up: Studying modified Lüscher approach

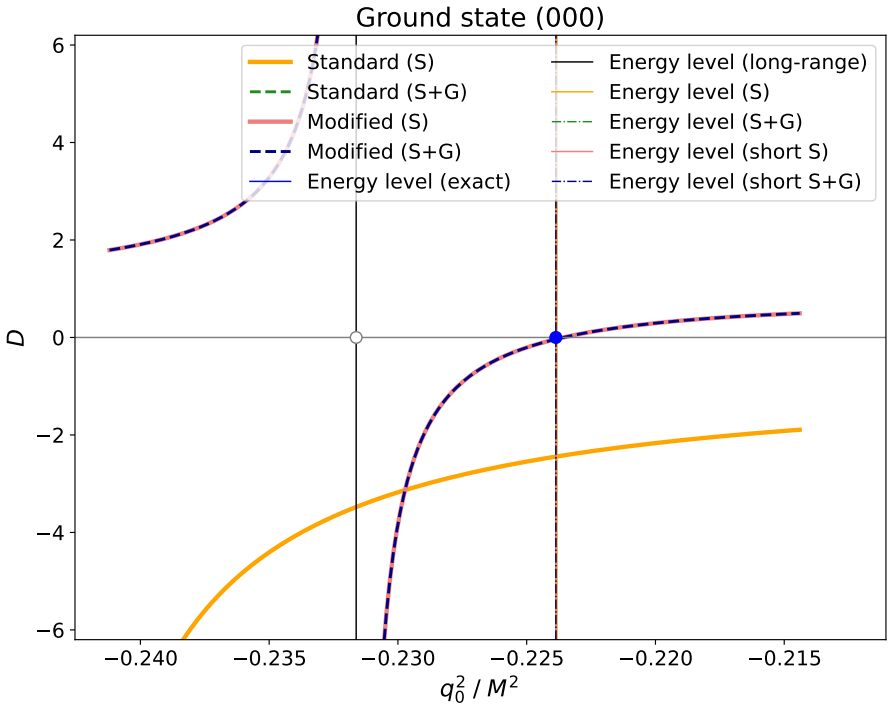
- Toy model:

$$V(\boldsymbol{p},\boldsymbol{q}) = \frac{4\pi g}{M^2 + (\boldsymbol{p}-\boldsymbol{q})^2} + \frac{4\pi g_S}{M_S^2 + (\boldsymbol{p}-\boldsymbol{q})^2}\,,$$

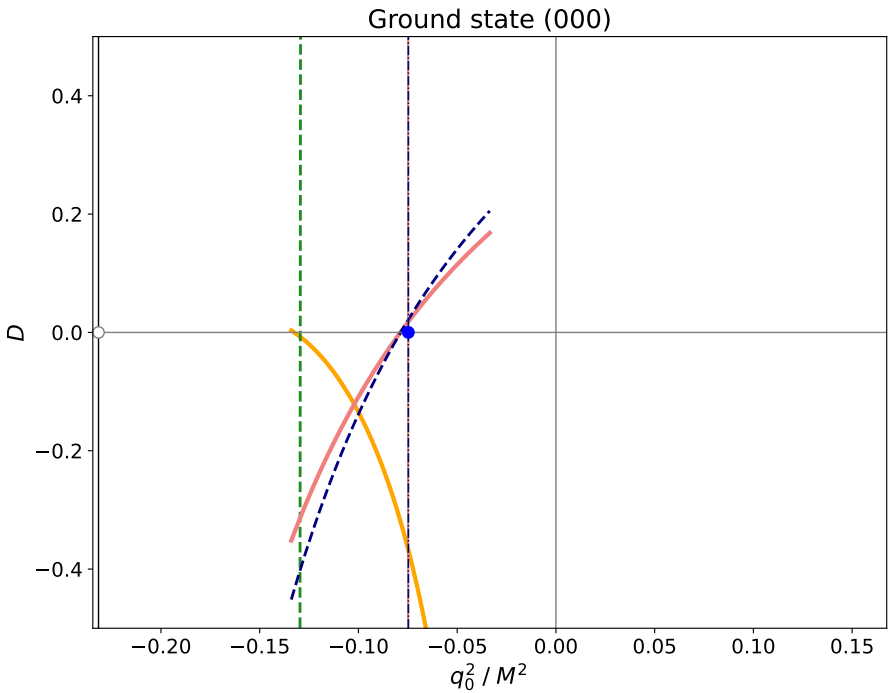
  with $g_S/g = -2.74$, $m/M = 6.7$ and M_S is taken to be $2M$ or $10M$.
- The finite-volume quantization condition with the above potential is solved, finding the eigenvalues of the Hamiltonian in the plane-wave basis to generate synthetic energy levels.
- The modified quantization condition is solved and the result is compared to the Hamiltonian energy levels.

Modified Quantization Condition

$M_s = 10M$

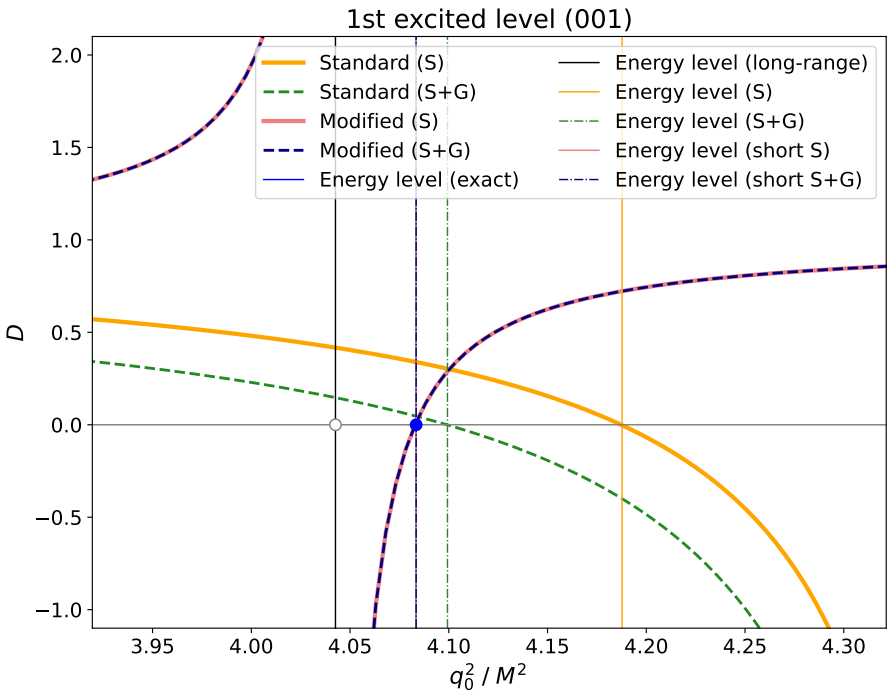


$M_s = 2M$

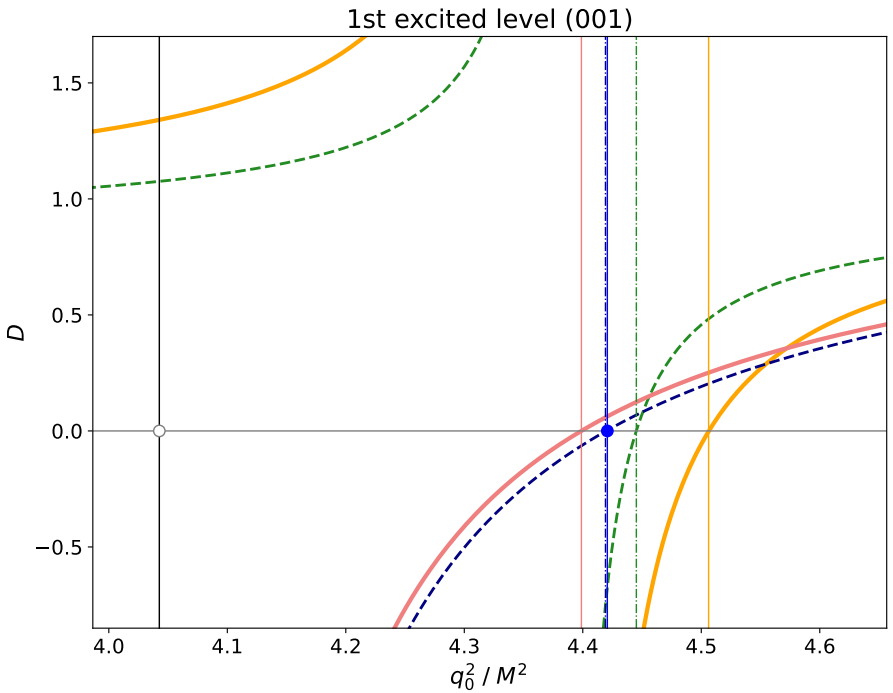


Modified Quantization Condition

$M_s = 10M$

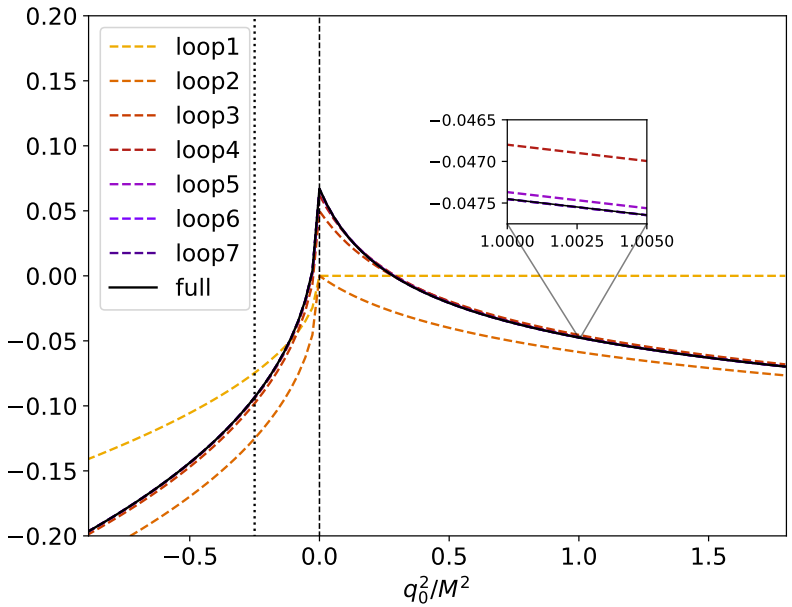


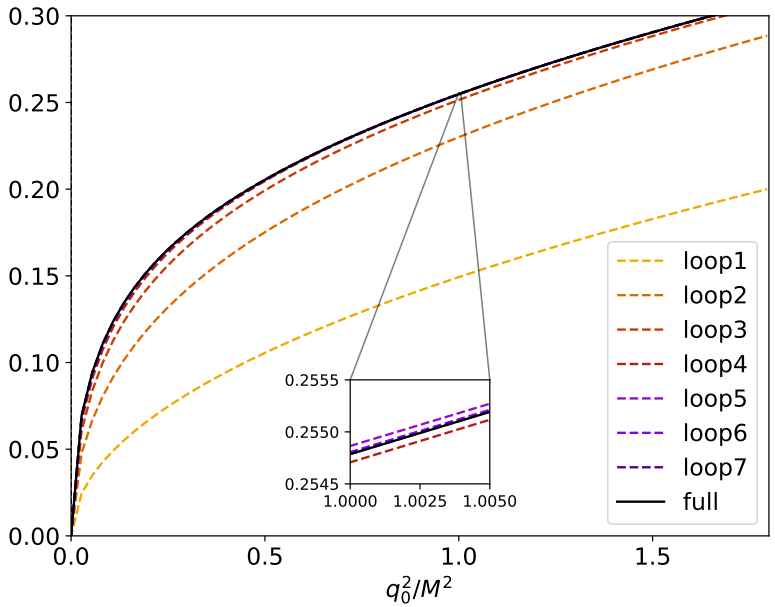
$M_s = 2M$



Loop function: $M_l(q)$

Real part (left) and Imaginary part (right) of $M_l(q)$ for $l = 0$:

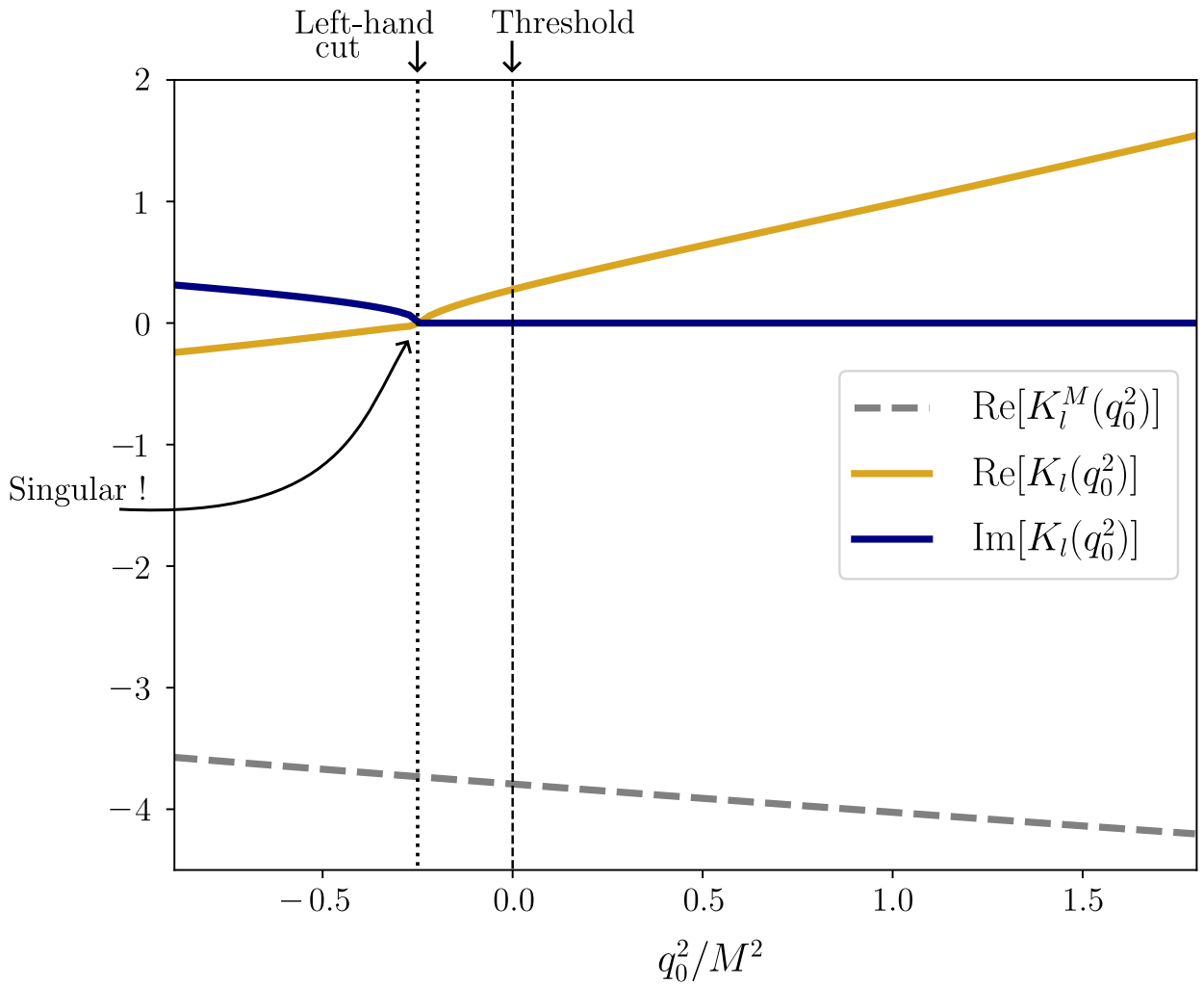




- Dimensional Regularization.
- Perturbative calculation using VEGAS algorithm.
- UV-convergent terms summed up to all orders, using Lippmann–Schwinger equation.

$K_l(q_0^2)$ vs. $K_l^M(q_0^2)$

Standard effective range function $K_l(q_0^2)$ (solid lines) vs. Modified effective range function $K_l^M(q_0^2)$ (dashed line):



QCD + QED with C* Boundary Conditions

Ming-Yang Duan (visiting student, 01.10.2024–31.03.2024) et al.

QCD + QED with C* boundary conditions

- QED with periodic boundary conditions is not a well-defined theory in a finite volume:

$$D_{\mu\nu}(\mathbf{x}, t) = \int \frac{dk_0}{2\pi} \frac{1}{L^3} \sum_{\mathbf{k}} \frac{e^{-ik_0 t + i\mathbf{k}\cdot\mathbf{x}}}{-k_0^2 - \mathbf{k}^2 - i\varepsilon} (\mathbf{k} = 0, \text{ zero mode}).$$

- **QED$_\mathbf{L}$**: excluding zero mode by using Faddeev–Popov procedure.
  - Leads to non-local theory.
  - Appelquist–Carazzone decoupling theorem does not work.
  - Relativistic/Non-relativistic EFT are not equivalent.
  - Implications for 2- and 3-body quantisation conditions?

- **QED$_C$**: Applying C-odd boundary condition.
  - Leads to a local theory.
  - Finite volume corrections to the 1-particle energy levels are the same in relativistic QFT and NREFT. (*M.-Y. Duan*)
  - Derivation of the 2- and 3-body quantisation conditions is in progress.
  - External collaborators: H.W. Hammer (Darmstadt), J.-J. Wu (Beijing), et al.

Summary

- Modified Lüscher equation in a background electromagnetic field has been derived.
- This allows one to extract the EM form factor of the ρ-meson on the lattice (in progress).
- Implementation of the modified Lüscher approach in the presence of long-range forces has been carried out (submitted to JHEP).
- The ρ-meson EM form factor in ChPT with an explicit ρ-meson field has been performed. Matching to NREFT is in progress.
- Implementing C* boundary condition in NREFT with an aim to derive the 2- and 3-body quantisation conditions (exploratory phase).